1. Import the required Python libraries.

```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import StandardScaler
from sklearn.preprocessing import LabelEncoder

# Ignore warnings
import warnings
warnings.filterwarnings("ignore")
```

2. Load the Telco Customer Churn Dataset into a Pandas DataFrame. And display First 10 records, Last 10 records, Shape of the dataset, Column names.

```
from google.colab import files
uploaded = files.upload()
```

Choose Files telco.csv

telco.csv(text/csv) - 1932822 bytes, last modified: 8/7/2026 - 100% done

```
Saving telco.csv to telco.csv
```

```
import pandas as pd

df = pd.read_csv("telco.csv")
```

```
df.head(10)
```

	Customer ID	Gender	Age	Under 30	Senior Citizen	Married	Dependents	Number of Dependents	Country	State	...
0	8779-QRDMV	Male	78	No	Yes	No	No	0	United States	California	...
1	7495-OOKFY	Female	74	No	Yes	Yes	Yes	1	United States	California	...
2	1658-BYGOY	Male	71	No	Yes	No	Yes	3	United States	California	...
3	4598-XLKNJ	Female	78	No	Yes	Yes	Yes	1	United States	California	...
4	4846-WHAFZ	Female	80	No	Yes	Yes	Yes	1	United States	California	...
5	4412-YLTKF	Female	72	No	Yes	No	Yes	1	United States	California	...
6	0390-DCFDQ	Female	76	No	Yes	Yes	Yes	2	United States	California	...
7	3445-HXXGF	Male	66	No	Yes	Yes	No	0	United States	California	...
8	2656-FMOKZ	Female	70	No	Yes	No	Yes	2	United States	California	...
9	2070-FNEXE	Female	77	No	Yes	No	Yes	2	United States	California	...

10 rows × 50 columns

```
df.tail(10)
```

	Customer ID	Gender	Age	Under 30	Senior Citizen	Married	Dependents	Number of Dependents	Country	State	..
7033	9281-CEDRU	Female	23	Yes	No	Yes	No	0	United States	California	
7034	0871-OPBXW	Female	57	No	No	No	No	0	United States	California	
7035	9767-FFLEM	Male	63	No	No	No	No	0	United States	California	
7036	8456-QDAVC	Male	57	No	No	No	No	0	United States	California	
7037	7750-EYXWZ	Female	62	No	No	No	No	0	United States	California	
7038	2569-WGERO	Female	30	No	No	No	No	0	United States	California	
7039	6840-RESVB	Male	38	No	No	Yes	Yes	2	United States	California	
7040	2234-XADUH	Female	30	No	No	Yes	Yes	2	United States	California	
7041	4801-JZAZL	Female	32	No	No	Yes	Yes	2	United States	California	
7042	3186-AJIEK	Male	44	No	No	No	No	0	United States	California	

10 rows × 50 columns

```
print(df.shape)
```

```
(7043, 50)
```

```
print(df.columns)
```

```
Index(['Customer ID', 'Gender', 'Age', 'Under 30', 'Senior Citizen', 'Married',
       'Dependents', 'Number of Dependents', 'Country', 'State', 'City',
       'Zip Code', 'Latitude', 'Longitude', 'Population', 'Quarter',
       'Referred a Friend', 'Number of Referrals', 'Tenure in Months', 'Offer',
       'Phone Service', 'Avg Monthly Long Distance Charges', 'Multiple Lines',
       'Internet Service', 'Internet Type', 'Avg Monthly GB Download',
       'Online Security', 'Online Backup', 'Device Protection Plan',
       'Premium Tech Support', 'Streaming TV', 'Streaming Movies',
       'Streaming Music', 'Unlimited Data', 'Contract', 'Paperless Billing',
       'Payment Method', 'Monthly Charge', 'Total Charges', 'Total Refunds',
       'Total Extra Data Charges', 'Total Long Distance Charges',
       'Total Revenue', 'Satisfaction Score', 'Customer Status', 'Churn Label',
       'Churn Score', 'CLTV', 'Churn Category', 'Churn Reason'],
      dtype='object')
```

3. Explore the Dataset and display Dataset information, Summary statistics, Number of missing values, Number of duplicate records

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 50 columns):
 #   Column                             Non-Null Count  Dtype
---  ------                             --------------  -----
 0   Customer ID                        7043 non-null   object
```

```
 1   Gender                             7043 non-null   object
 2   Age                                7043 non-null   int64
 3   Under 30                           7043 non-null   object
 4   Senior Citizen                     7043 non-null   object
 5   Married                            7043 non-null   object
 6   Dependents                         7043 non-null   object
 7   Number of Dependents               7043 non-null   int64
 8   Country                            7043 non-null   object
 9   State                              7043 non-null   object
 10  City                               7043 non-null   object
 11  Zip Code                           7043 non-null   int64
 12  Latitude                           7043 non-null   float64
 13  Longitude                          7043 non-null   float64
 14  Population                         7043 non-null   int64
 15  Quarter                            7043 non-null   object
 16  Referred a Friend                  7043 non-null   object
 17  Number of Referrals                7043 non-null   int64
 18  Tenure in Months                   7043 non-null   int64
 19  Offer                              3166 non-null   object
 20  Phone Service                      7043 non-null   object
 21  Avg Monthly Long Distance Charges  7043 non-null   float64
 22  Multiple Lines                     7043 non-null   object
 23  Internet Service                   7043 non-null   object
 24  Internet Type                      5517 non-null   object
 25  Avg Monthly GB Download            7043 non-null   int64
 26  Online Security                    7043 non-null   object
 27  Online Backup                      7043 non-null   object
 28  Device Protection Plan             7043 non-null   object
 29  Premium Tech Support               7043 non-null   object
 30  Streaming TV                       7043 non-null   object
 31  Streaming Movies                   7043 non-null   object
 32  Streaming Music                    7043 non-null   object
 33  Unlimited Data                     7043 non-null   object
 34  Contract                           7043 non-null   object
 35  Paperless Billing                  7043 non-null   object
 36  Payment Method                     7043 non-null   object
 37  Monthly Charge                     7043 non-null   float64
 38  Total Charges                      7043 non-null   float64
 39  Total Refunds                      7043 non-null   float64
 40  Total Extra Data Charges           7043 non-null   int64
 41  Total Long Distance Charges        7043 non-null   float64
 42  Total Revenue                      7043 non-null   float64
 43  Satisfaction Score                 7043 non-null   int64
 44  Customer Status                    7043 non-null   object
 45  Churn Label                        7043 non-null   object
 46  Churn Score                        7043 non-null   int64
 47  CLTV                               7043 non-null   int64
 48  Churn Category                     1869 non-null   object
 49  Churn Reason                       1869 non-null   object
dtypes: float64(8), int64(11), object(31)
memory usage: 2.7+ MB
```

```
df.describe()
```

	Age	Number of Dependents	Zip Code	Latitude	Longitude	Population	Number of Referrals
count	7043.000000	7043.000000	7043.000000	7043.000000	7043.000000	7043.000000	7043.000000
mean	46.509726	0.468692	93486.070567	36.197455	-119.756684	22139.603294	1.951867
std	16.750352	0.962802	1856.767505	2.468929	2.154425	21152.392837	3.001199
min	19.000000	0.000000	90001.000000	32.555828	-124.301372	11.000000	0.000000
25%	32.000000	0.000000	92101.000000	33.990646	-121.788090	2344.000000	0.000000
50%	46.000000	0.000000	93518.000000	36.205465	-119.595293	17554.000000	0.000000
75%	60.000000	0.000000	95329.000000	38.161321	-117.969795	36125.000000	3.000000
max	80.000000	9.000000	96150.000000	41.962127	-114.192901	105285.000000	11.000000

```
df.describe(include='all')
```

	Customer ID	Gender	Age	Under 30	Senior Citizen	Married	Dependents	Number of Dependents	Country
count	7043	7043	7043.000000	7043	7043	7043	7043	7043.000000	7043
unique	7043	2	NaN	2	2	2	2	NaN	1
top	3186-AJIEK	Male	NaN	No	No	No	No	NaN	United States
freq	1	3555	NaN	5642	5901	3641	5416	NaN	7043
mean	NaN	NaN	46.509726	NaN	NaN	NaN	NaN	0.468692	NaN
std	NaN	NaN	16.750352	NaN	NaN	NaN	NaN	0.962802	NaN
min	NaN	NaN	19.000000	NaN	NaN	NaN	NaN	0.000000	NaN
25%	NaN	NaN	32.000000	NaN	NaN	NaN	NaN	0.000000	NaN
50%	NaN	NaN	46.000000	NaN	NaN	NaN	NaN	0.000000	NaN
75%	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	0.000000	NaN
max	NaN	NaN	80.000000	NaN	NaN	NaN	NaN	9.000000	NaN

11 rows × 50 columns

```
df.isnull().sum()
```

	0
Customer ID	0
Gender	0
Age	0
Under 30	0
Senior Citizen	0
Married	0
Dependents	0
Number of Dependents	0
Country	0
State	0
City	0
Latitude	0
Longitude	0
Population	0
Quarter	0
Number of Referrals	0
Tenure in Months	0
Offer	3877
Phone Service	0
Avg Monthly Long Distance Charges	0
Multiple Lines	0
Internet Service	0
Internet Type	1526
Avg Monthly GB Download	0
Online Security	0
Online Backup	0
Device Protection Plan	0
Premium Tech Support	0
Streaming TV	0
Streaming Movies	0
Streaming Music	0
Unlimited Data	0
Contract	0
Paperless Billing	0
Payment Method	0
Monthly Charge	0
Total Charges	0
Total Refunds	0

```
print(df.duplicated().sum())
```

```
0
```

4. Clean the data a. Remove duplicate customer records. b. Replace missing values i. Numerical columns → Median ii. Categorical columns → Mode c. Verify that no missing values remain.

```
df = df.drop_duplicates()
```

```
print(df.duplicated().sum())
```

```
0
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 50 columns):
 #   Column                             Non-Null Count  Dtype
---  ------                             --------------  -----
 0   Customer ID                        7043 non-null   object
 1   Gender                             7043 non-null   object
 2   Age                                7043 non-null   int64
 3   Under 30                           7043 non-null   object
 4   Senior Citizen                     7043 non-null   object
 5   Married                            7043 non-null   object
 6   Dependents                         7043 non-null   object
 7   Number of Dependents               7043 non-null   int64
 8   Country                            7043 non-null   object
 9   State                              7043 non-null   object
 10  City                               7043 non-null   object
 11  Zip Code                           7043 non-null   int64
 12  Latitude                           7043 non-null   float64
 13  Longitude                          7043 non-null   float64
 14  Population                         7043 non-null   int64
 15  Quarter                            7043 non-null   object
 16  Referred a Friend                  7043 non-null   object
 17  Number of Referrals                7043 non-null   int64
 18  Tenure in Months                   7043 non-null   int64
 19  Offer                              3166 non-null   object
 20  Phone Service                      7043 non-null   object
 21  Avg Monthly Long Distance Charges  7043 non-null   float64
 22  Multiple Lines                     7043 non-null   object
 23  Internet Service                   7043 non-null   object
 24  Internet Type                      5517 non-null   object
 25  Avg Monthly GB Download            7043 non-null   int64
 26  Online Security                    7043 non-null   object
```

```
 27  Online Backup                      7043 non-null   object
 28  Device Protection Plan             7043 non-null   object
 29  Premium Tech Support               7043 non-null   object
 30  Streaming TV                       7043 non-null   object
 31  Streaming Movies                   7043 non-null   object
 32  Streaming Music                    7043 non-null   object
 33  Unlimited Data                     7043 non-null   object
 34  Contract                           7043 non-null   object
 35  Paperless Billing                  7043 non-null   object
 36  Payment Method                     7043 non-null   object
 37  Monthly Charge                     7043 non-null   float64
 38  Total Charges                      7043 non-null   float64
 39  Total Refunds                      7043 non-null   float64
 40  Total Extra Data Charges           7043 non-null   int64
 41  Total Long Distance Charges        7043 non-null   float64
 42  Total Revenue                      7043 non-null   float64
 43  Satisfaction Score                 7043 non-null   int64
 44  Customer Status                    7043 non-null   object
 45  Churn Label                        7043 non-null   object
 46  Churn Score                        7043 non-null   int64
 47  CLTV                               7043 non-null   int64
 48  Churn Category                     1869 non-null   object
 49  Churn Reason                       1869 non-null   object
dtypes: float64(8), int64(11), object(31)
memory usage: 2.7+ MB
```

```
df.isnull().sum()
```

```
Total Extra Data Charges       0
Total Long Distance Charges    0
Total Revenue                  0
Satisfaction Score             0
Customer Status                0
Churn Label                    0
Churn Score                    0
CLTV                           0
Churn Category              5174
Churn Reason                5174
dtype: int64
```

	0
Customer ID	0
Gender	0
Age	0
Under 30	0
Senior Citizen	0
Married	0
Dependents	0
Number of Dependents	0
Country	0
State	0
City	0
Latitude	0
Longitude	0
Population	0
Quarter	0
Referred a Friend	0
Number of Referrals	0
Phone Service	0
Avg Monthly Long Distance Charges	0
Multiple Lines	0
Streaming Music	0
Contract	0
Paperless Billing	0
Payment Method	0
Monthly Charge	0
Total Charges	0
Total Refunds	0

```
categorical_cols = df.select_dtypes(include="object").columns
print(categorical_cols)
```

```
Index(['Customer ID', 'Gender', 'Under 30', 'Senior Citizen', 'Married',
       'Dependents', 'Country', 'State', 'City', 'Quarter',
       'Referred a Friend', 'Offer', 'Phone Service', 'Multiple Lines',
       'Internet Service', 'Internet Type', 'Online Security', 'Online Backup',
       'Device Protection Plan', 'Premium Tech Support', 'Streaming TV',
       'Streaming Movies', 'Streaming Music', 'Unlimited Data', 'Contract',
       'Paperless Billing', 'Payment Method', 'Customer Status', 'Churn Label',
       'Churn Category', 'Churn Reason'],
      dtype='object')
```

```
for col in categorical_cols:
    df[col].fillna(df[col].mode()[0], inplace=True)
```

5. Encode categorical columns: Gender, Partner, Dependents, Contract, PaymentMethod, InternetService, Churn. Use appropriate encoding techniques.

```
le = LabelEncoder()

columns = [
    "Gender",
    "Married", # Corrected from "Partner"
    "Dependents",
    "Contract",
    "Payment Method",
    "Internet Service",
    "Churn Label"
]

for col in columns:
    df[col] = le.fit_transform(df[col])
```

```
df.head()
```

	Customer ID	Gender	Age	Under 30	Senior Citizen	Married	Dependents	Number of Dependents	Country	State	...
0	8779-QRDMV	1	78	No	Yes	0	0	0	United States	California	...
1	7495-OOKFY	0	74	No	Yes	1	1	1	United States	California	...
2	1658-BYGOY	1	71	No	Yes	0	1	3	United States	California	...
3	4598-XLKNJ	0	78	No	Yes	1	1	1	United States	California	...
4	4846-WHAFZ	0	80	No	Yes	1	1	1	United States	California	...

5 rows × 50 columns

Total Extra Data Charges 0
Total Long Distance Charges 0
Total Revenue 0
Churn Label 0
Churn Score 0
Churn Reason 5174
dtype: int64

6. Standardise the numerical features: Tenure, MonthlyCharges, TotalCharges, using StandardScaler.

```
scaler = StandardScaler()

df[["Tenure in Months","Monthly Charge","Total Charges"]] = scaler.fit_transform(
    df[["Tenure in Months","Monthly Charge","Total Charges"]]
)
df.head()
```

	Customer ID	Gender	Age	Under 30	Senior Citizen	Married	Dependents	Number of Dependents	Country	State	...
0	8779-QRDMV	1	78	No	Yes	0	0	0	United States	California	...
1	7495-OOKFY	0	74	No	Yes	1	1	1	United States	California	...
2	1658-BYGOY	1	71	No	Yes	0	1	3	United States	California	...
3	4598-XLKNJ	0	78	No	Yes	1	1	1	United States	California	...
4	4846-WHAFZ	0	80	No	Yes	1	1	1	United States	California	...

5 rows × 50 columns

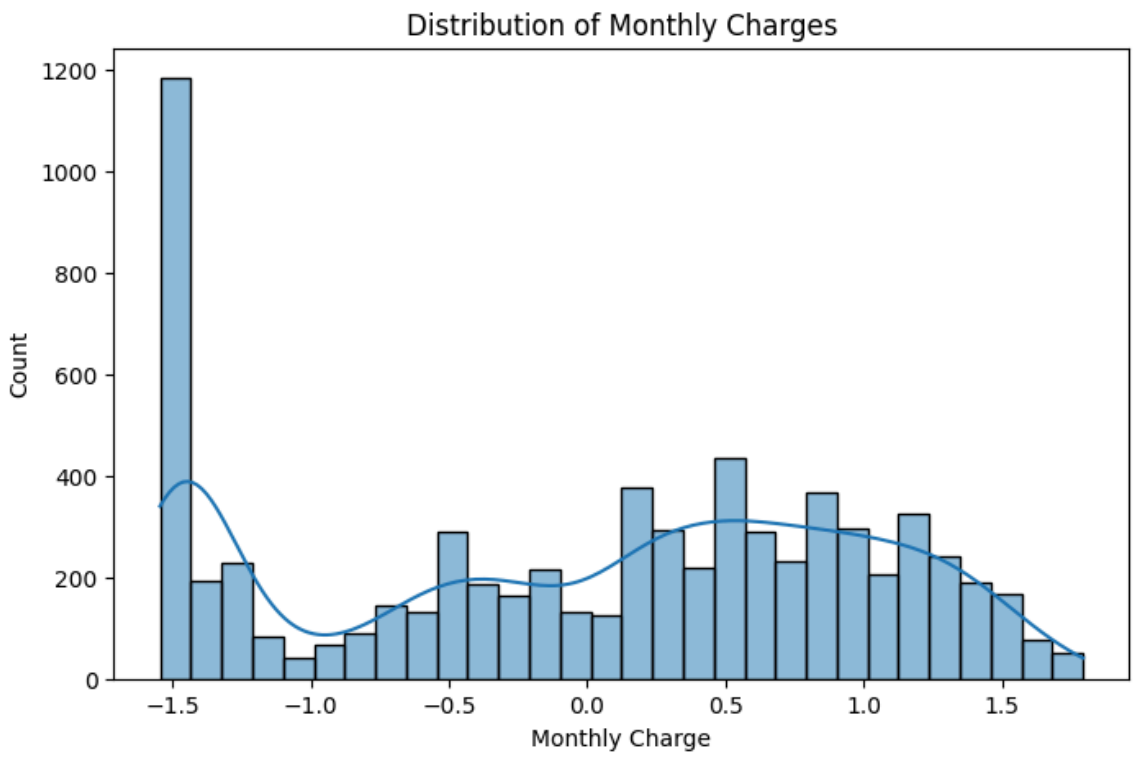
7. Plot Distribution of MonthlyCharges, Distribution of Tenure.

```
plt.figure(figsize=(8,5))

sns.histplot(df["Monthly Charge"], bins=30, kde=True)
```

```
plt.title("Distribution of Monthly Charges")

plt.show()
```

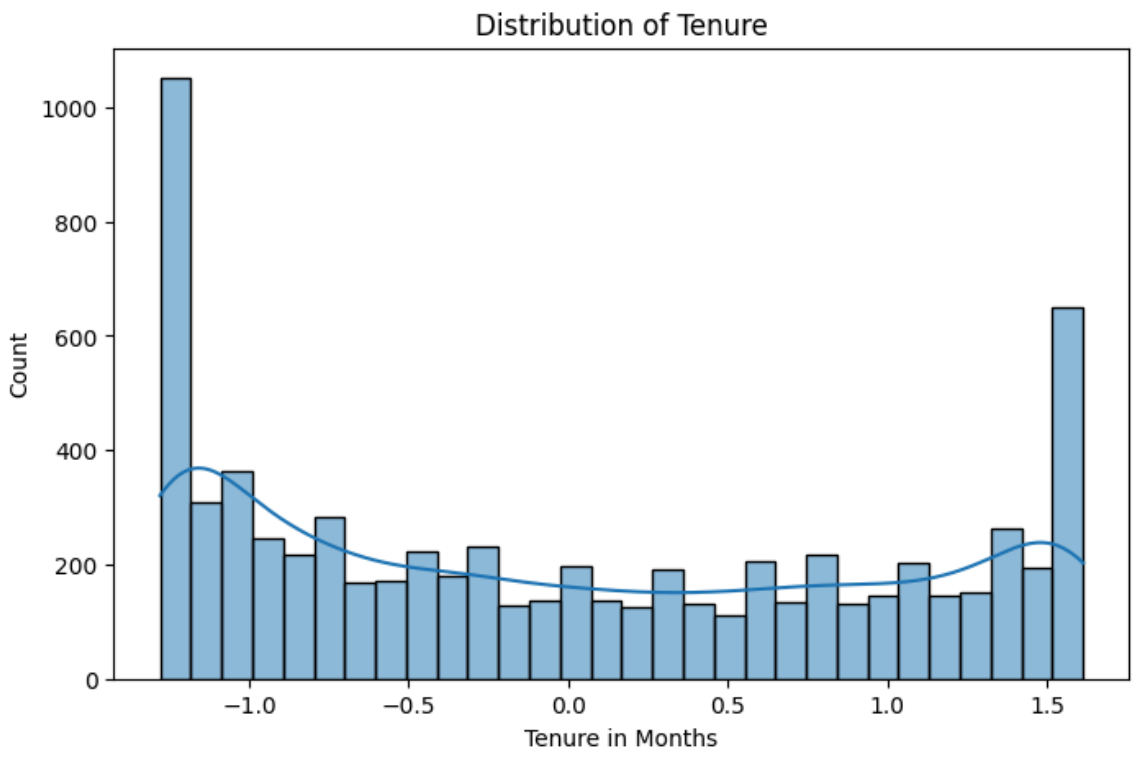



```
plt.figure(figsize=(8,5))

sns.histplot(df["Tenure in Months"], bins=30, kde=True)

plt.title("Distribution of Tenure")

plt.show()
```



8. Create a Count Plot showing Contract vs Churn.